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Power transmission system

Abstract

A power transmission assembly includes a base, a rotating frame rotatably coupled to the base and configured to rotate about an axis, an output shaft coupled to the rotating frame, a weight selectively repositionable relative to the rotating frame, and a weight actuator configured to reposition the weight relative to the rotating frame to move the weight from a subtraction position located at a first height to an addition position located at a second height. The second height is greater than the first height such that a gravitational force on the weight drives the rotating frame to rotate about the axis and drive the output shaft.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/437,613, filed Sep. 9, 2021, which is a 371

national stage of International Patent Application No. PCT/US2021/033426, filed May 20, 2021, which claims the benefit of U.S. Provisional Patent Application No. 63/031,126, filed May 28, 2020, all of which are incorporated herein by reference in their entireties.

BACKGROUND

(1) The present disclosure relates generally to power transmissions. More specifically, the present disclosure relates to power transmissions that receive a first type of mechanical energy (e.g., linear movement of an object) as an input and output a second type of mechanical energy (e.g., rotational mechanical energy).

SUMMARY

(2) At least one embodiment relates to a power transmission assembly including a base, a rotating frame rotatably coupled to the base and configured to rotate about an axis, an output shaft coupled to the rotating frame, a weight selectively repositionable relative to the rotating frame, and a weight actuator configured to reposition the weight relative to the rotating frame to move the weight from a subtraction position located at a first height to an addition position located at a second height. The second height is greater than the first height such that a gravitational force on the weight drives the rotating frame to rotate about the axis and drive the output shaft.

(3) This summary is illustrative only and is not intended to be in any way limiting. Other aspects, inventive features, and advantages of the devices or processes described herein will become apparent in the detailed description set forth herein, taken in conjunction with the accompanying figures, wherein like reference numerals refer to like elements.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) FIG. 1 is a front view of a power transmission system, according to various exemplary embodiments.

(2) FIG. 2 is a right side view of the power transmission system of FIG. 1.

(3) FIG. 3 is front view of the power transmission system of FIG. 1 in a first position.

(4) FIG. 4 is front view of the power transmission system of FIG. 1 in a second position.

(5) FIG. 5 is front view of the power transmission system of FIG. 1 in a third position.

(6) FIG. 6 is front view of the power transmission system of FIG. 1 in a fourth position.

(7) FIG. 7 is a block diagram of a control system for the power transmission system of FIG. 1, according to an exemplary embodiment.

(8) FIG. 8 is a top view of a system including several of the power transmission systems of FIG. 1, according to an exemplary embodiment.

(9) FIG. 9 is a front view the power transmission system of FIG. 1, according to an exemplary embodiment.

(10) FIG. 10 is a front view the power transmission system of FIG. 1, according to another exemplary embodiment.

(11) FIG. 11 is a front view the power transmission system of FIG. 1, according to another exemplary embodiment.

(12) FIG. 12 is a top view of a power transmission system, according to an exemplary embodiment.

(13) FIG. 13 is a front view of the power transmission system of FIG. 12.

DETAILED DESCRIPTION

(14) Before turning to the figures, which illustrate certain exemplary embodiments in detail, it should be understood that the present disclosure is not limited to the details or methodology set forth in the description or illustrated in the figures. It should also be understood that the terminology used herein is for the purpose of description only and should not be regarded as limiting.

(15) Referring generally to the figures, a power transmission is shown according to various exemplary embodiments. The power transmission includes a rotating frame or wheel that is rotatably coupled to a base. The wheel rotates about a substantially horizontal axis of rotation. In some embodiments, the wheel is coupled to an electromagnetic device such that rotation of the wheel drives the electromagnetic device to generate electrical energy. One or more weights are selectively repositionable relative to the wheel. Specifically, the weights are moved relative to the wheel by a weight actuator to drive rotation of the wheel.

(16) According to one embodiment, the power transmission operates according to a cycle. At the beginning of the cycle, the weight is (a) coupled to the wheel, (b) offset a distance from the axis of rotation, and (c) located at an addition position relative to the wheel. The addition position is located at a first, elevated height. Gravity imparts a downward force on the weight. Due to the offset distance of the weight from the axis of rotation, the force of gravity imparts a moment load on the wheel, driving the wheel to rotate about the axis of rotation. As the wheel rotates, the weight moves downward, toward a subtraction position having a second height that is lower than the first height. When the wheel has rotated sufficiently for the weight to have reached the subtraction position, the weight actuator engages the weight, moving the weight relative to the wheel. The weight actuator returns the weight to the addition position where the weight is again coupled to the wheel, completing the cycle of operation. The cycle may be repeated to continue rotation of the wheel.

(17) Referring to FIGS. 1 and 2, a power transmission system, wheel assembly, power transmission assembly, power transmission, power conversion assembly, or generator, is shown as system **10** according to an exemplary embodiment. The system **10** includes a chassis, frame, or base assembly, shown as base **20**. The base **20** is configured to support the various components of the system **10**. The base **20** includes a bottom portion, base portion, or horizontal portion, shown as foot **22** and a top portion, upper portion, or vertical portion, shown as support wall **24**, that is fixedly coupled to the foot **22** and extends upward from the foot **22**. The support wall **24** defines an aperture or passage, shown as output shaft passage **26**, that extends through the support wall **24**. The shaft passage **26** may be at least partially defined by a bushing or bearing that facilitates rotation of a shaft extending through the shaft passage **26** relative to the support wall **24**.

(18) The system **10** further includes a wheel, rotating frame, or rotating support structure, shown as wheel **30**. The wheel **30** is fixedly coupled to a shaft, shown as output shaft **32**. The output shaft **32** extends through the shaft passage **26**, such that the output shaft **32** rotatably couples the wheel **30** to the base **20**. The wheel **30** rotates about an axis of rotation, shown as axis **34**. The axis **34** extends along the output shaft **32** (e.g., is centered about the output shaft **32**). In some embodiments, the axis **34** extends substantially horizontally. In such embodiments, the wheel **30** rotates within a substantially vertical plane (e.g., a plane perpendicular to the horizontal axis **34**).

(19) The system **10** includes one or more weights **40** that are coupled to the wheel **30**. Specifically, in the embodiment shown in FIG. 1, the weights **40** include a weight **40A**, a weight **40B**, a weight **40C**, and a weight **40D**. Each of the weights **40** may have the same mass, or one or more of the weights **40** may have different masses. Each weight **40** has a center of gravity **C** that is offset from a distance **D** from the axis **34**. As shown in FIG. 1, the distance **D** is equal for each of the weights **40**. In other embodiments, the weights **40** are positioned at different distances from the axis **34** such that the distance **D** varies for one or more of the weights **40**. The center of gravity **C** of each weight **40** is positioned at a vertical position or height **H**. As shown, each height is measured relative to the bottom of the base **20**. In other embodiments, the height **H** is measured from another location. The center of gravity **C** of each weight **40** is located at an angular position Θ that corresponds to the height **H** of the center of gravity **C**. As shown, the angular position Θ is 0 degrees when the corresponding height **H** of the weight **40** is maximized (e.g., at top dead center), and the angular position Θ is 180 degrees when the corresponding height **H** of the weight **40** is minimized (e.g., at bottom dead center). In other embodiments, the angular position Θ of each weight **40** is otherwise

measured.

(20) In some embodiments, the weights **40** are selectively repositionable relative to the wheel **30**. In some embodiments, the weights **40** are selectively coupled to the wheel **30** such that the weights **40** are decoupled from the wheel **30** when the weights **40** are repositioned relative to the wheel **30**. By way of example, the weights **40** may be decoupled (e.g., by disconnecting a latch or other mechanical coupler, by disconnecting the power supply to an electromagnet, etc.). In other embodiments, the weights **40** are slidably coupled to the wheel **30** such that the weights remain coupled to the wheel **30** while the weights **40** are repositioned relative to the wheel **30**. By way of example, the weights **40** may be slidably coupled to a track positioned on the wheel **30** such that the weights **40** travel along the track.

(21) Referring to FIG. 2, the system **10** includes an actuator, shown as weight actuator **50**, that is configured to reposition the weights **40** relative to the wheel **30**. In some embodiments, the weight actuator **50** includes a weight interface or end effector, shown as interface **52**, and a height actuator or weight shifter, shown as lift **54**. The interface **52** is configured to couple (e.g., selectively, continuously, permanently, etc.) the weights **40** to the weight actuator **50**. With the weights **40** coupled to the interface **52**, the lift **54** is configured provide mechanical energy to raise the weights **40** relative to the wheel **30**. Accordingly, the weight actuator **50** is configured to lift or raise the weights **40** from a first, low elevation to a second, greater elevation. By way of example, the weight actuator **50** may raise the weight **40A** from a height $H_{sub.1}$ of 1 ft to a height $H_{sub.1}$ of 5 ft.

(22) Referring to FIGS. 3-6, an exemplary portion of a cycle of operation of the system **10** is shown according to an exemplary embodiment. In FIG. 3, the weights **40** are all coupled to the wheel **30**. The weights **40** are all positioned on the wheel **30** such that the centers of gravity **C** of each of the weights **40** are all located on one side of the wheel (e.g., $0^\circ < \Theta < 180^\circ$). Gravity applies a downward gravitational force **G** onto each weight **40** at the corresponding center of gravity **C**. Due to the separation of the centers of gravity **C** from the axis **34**, the gravitational force **G** on each weight **40** imparts a moment loading on the wheel **30**, driving the wheel **30** to rotate in a first direction (e.g., counterclockwise as shown in FIG. 3). As the wheel **30** rotates, the horizontal distance between each center of gravity **C** and the axis **34** (i.e., the length of the moment arm) varies, varying the moment load imparted by each weight **40**. The moment load **M** exerted by a given weight **40** can be calculated using the following equation:

$$M = GD \sin \theta = mgD \sin \theta \quad (1)$$

where **G** is the gravitational force on a given weight **40**, **D** is the distance of the weight **40** from the axis **34**, Θ is the angular position of the weight **40**, **m** is the mass of the weight **40**, and **g** is the acceleration due to gravity experienced by the weight **40** (e.g., 9.81 m/s²). The moment effect of each weight **40** can be calculated separately and combined to determine the total moment load of the weights **40** on the wheel **30**.

(23) Because each of the weights **40** are located on the left side of the wheel **30**, the sum of the moment loads of the weights **40** is positive, and the wheel **30** is driven to rotate in the first direction. As the wheel **30** rotates, the weights **40** all move downward. As shown in FIG. 4, the weight **40D** eventually reaches a position or range of positions (e.g., a bottom position or subtraction position), shown as subtraction position **60**. The subtraction position **60** is fixed in space (e.g., relative to the base **20**). In some embodiments, the subtraction position **60** corresponds to a bottom dead center position of the weight **40D** (i.e., where the angular position Θ of the weight **40D** is 180 degrees), such that the height **H** of the weight **40D** would begin increasing if the wheel **30** rotated further in the first direction. In other embodiments, the subtraction position corresponds to a position of the weight **40D** above bottom dead center (e.g., where the angular position Θ of the weight **40D** is less than 180 degrees).

(24) When the weight **40D** reaches the subtraction position **60**, the interface **52** engages the weight **40D**, and the lift **54** applies an upward force to the weight **40D**, lifting the weight **40D**. The weight

actuator **50** may lift the weight **40D** straight upward or along a path that moves upward and laterally (e.g., to avoid one or more obstacles). As shown in FIG. 5, the weight actuator **50** moves the weight **40D** to an intermediate position **62**. In some embodiments, the weight **40D** is slidably coupled to the wheel **30** such that the weight **40D** remains coupled to the wheel **30** while in the intermediate position **62**. In other embodiments, the weight **40D** is selectively coupled to the wheel **30**, and the weight **40D** is decoupled from and/or separated from the wheel **30** while in the intermediate position **62**.

(25) As shown in FIG. 6, the weight actuator **50** moves the weight **40D** to a position or range of positions (e.g., a top position or addition position), shown as addition position **64**. The addition position **64** is fixed in space (e.g., relative to the base **20**). In some embodiments, the addition position **64** corresponds to a top dead center position of the weight **40D** (i.e., where the angular position Θ of the weight **40D** is 0 degrees), such that the height H of the weight **40D** would begin decreasing if the wheel **30** rotated further in the first direction. In other embodiments, the addition position **64** corresponds to a position of the weight **40D** below top dead center (e.g., where the angular position Θ of the weight **40D** is greater than 0 degrees).

(26) When the weight **40D** reaches the addition position **64**, the weight **40D** is again coupled (e.g., fixedly coupled, rotatably coupled, etc.) to the wheel **30** such that the weight **40D** is again positioned at the distance D from the axis **34**. With the weight **40D** coupled to the wheel **30**, the weight **40D** again moves with the wheel **30**. As the wheel **30** continues to rotate, the weight **40D** moves downward until the weight **40D** again reaches the subtraction position **60**, and the cycle of operation shown in FIGS. 3-6 may be repeated to facilitate continuous rotation of the wheel **30**.

(27) The system **10** may also control the weight actuator **50** to lift the other weights (e.g., weights **40A**, **40B**, and **40C**) to the addition position **64** in response to the weight reaching the subtraction position **60**. By way of example, when the weight **40C** reaches the subtraction position **60**, the weight actuator **50** may engage the weight **40C** and lift the weight **40C** to the addition position **64** through the intermediate position **62**. By way of another example, when the weight **40B** reaches the subtraction position **60**, the weight actuator **50** may engage the weight **40B** and lift the weight **40B** to the addition position **64** through the intermediate position **62**. By way of another example, when the weight **40A** reaches the subtraction position **60**, the weight actuator **50** may engage the weight **40A** and lift the weight **40A** to the addition position **64** through the intermediate position **62**. In this way, the system **10** maintains a cycle of operation that continuously replaces each weight **40** at an elevated position such that each weight **40** continues to drive the wheel **30** over time.

(28) By utilizing multiple repositionable weights **40**, the moment load on the wheel **30** may be increased, increasing the output torque that the system **10** is capable of delivering. Additionally, the use of multiple weights **40** facilitates applying a more consistent torque to the wheel **30**. As the wheel **30** rotates, the moment load imparted by each weight **40** varies both (a) due to the variation in the length of the moment arm of the weight **40** as the wheel **30** rotates and (b) due to the reduction in torque when the weight **40** is being moved from the subtraction position **60** to the addition position **64**. Adding more weights **40** to the system **10** reduces the effect of these variations in torque on the overall torque experienced by the wheel **30**, as the torque provided by each weight **40** then makes up a lesser portion of the overall torque. By way of example, in a system **10** that includes only the weight **40D**, the wheel **30** may experience no torque when the weight **40D** is being moved from the subtraction position **60** to the addition position **64**. However, in a system **10** that includes both the weight **40B** and the weight **40D**, the weight **40B** continues to impart a moment load on the wheel **30** while the weight **40D** is being moved from the subtraction position **60** to the addition position **64**.

(29) Referring again to FIG. 2, the torque imparted on the wheel **30** by the weights **40** is transferred by the output shaft **32**. As shown in FIG. 2, the output shaft **32** is coupled to a power take off (PTO) output, shown as PTO **70**. Specifically, the output shaft **32** is shown as being directly coupled to the PTO **70** such that the output shaft **32** drives the PTO **70**. The output shaft **32** is coupled to an

electromagnetic device or motor/generator, shown as electric motor **72**. Specifically, a shaft of the electric motor **72** is coupled to the output shaft **32** by a power transmission (e.g., a gearbox, a chain and sprocket assembly, a belt and pulley assembly, a gear train, etc.), shown as belt **74**. In other embodiments, the electric motor **72** is directly coupled to the output shaft **32**. In other embodiments, the PTO **70** is coupled to the output shaft **32** by the belt **74** or another type of power transmission. In other embodiments, the PTO **70** and/or the electric motor **72** are omitted.

(30) The PTO **70** may include one or more devices that receive and utilize rotational mechanical energy from the output shaft **32**. The PTO **70** may include one or more devices that convert rotational mechanical energy to another form. By way of example, the PTO **70** may include a pump or compressor that receives rotational mechanical energy and provides a flow of a pressurized fluid (e.g., a gas, a liquid, etc.). By way of another example, the PTO **70** may include a generator that receives rotational mechanical energy and provides electrical energy. The PTO **70** may include one or more devices that utilize rotational mechanical energy to perform one or more functions. By way of example, the PTO **70** may include a wheel (e.g., as part of a drivetrain for a vehicle), a conveyor, an implement (e.g., a saw, a lathe, a mill, a washing machine, etc.), or another type of device.

(31) The electric motor **72** may be configured to receive rotational mechanical energy and provide electrical energy. The electric motor **72** may provide alternating current or direct current electrical energy. The electric motor **72** may be configured to receive electrical energy and provide rotational mechanical energy. By way of example, the electric motor **72** may drive the wheel **30** and/or the PTO **70**. Electrical energy provided (e.g., generated) by the electric motor **72** may be used to power any functions of the system **10**.

(32) In some embodiments, the system **10** includes one or more energy storage devices, shown as batteries **76**. The batteries **76** may receive and store electrical energy provided by the electric motor **72**. The batteries **76** may provide stored electrical energy to power the electric motor **72**. The batteries **76** may include one or more lithium ion batteries, lead acid batteries, nickel metal hydride batteries, or other types of batteries. Additionally or alternatively, the system **10** may include another type of energy storage device, such as one or more capacitors.

(33) In some embodiments, the system **10** includes one or more electrical loads, shown as loads **78**. The loads **78** may be electrically coupled to the electric motor **72** and/or the batteries **76**. The loads **78** may be configured to consume electrical energy from the electric motor **72** and/or the batteries **76**. The loads **78** may additionally or alternatively be configured to provide electrical energy to the electric motor and/or the batteries **76**. By way of example, the loads **78** may include electric motors, heating elements, electronics (e.g., controllers, displays, etc.), or other electrical loads. By way of another example, the loads **78** may include an electrical grid that consumes electrical energy from the system **10** and/or provides electrical energy to the system **10**.

(34) In some embodiments, it is advantageous to minimize the weight of the wheel **30** and maximize the weight of the weights **40**. This arrangement maximizes the amount of output power provided by the system **10** for weights of a given size while minimizing the energy required to accelerate the wheel **30**. In some embodiments, the wheel **30** includes relatively lightweight materials, such as aluminum or bamboo. In some embodiments, the weights **40** include relatively dense materials, such as lead, mercury, steel, or iron.

(35) The path along which the weights **40** move between the subtraction position **60** and the addition position **64** may vary between different embodiments. In some embodiments, the weights **40** move straight upward, along the side of the wheel **40** or through the center of the wheel. In some embodiments, the path of the weights **40** curves such that the momentum of the weights **40** caused by spinning of the wheel **30** carries the weights **40** along the path. In some embodiments, the momentum of the weights **40** may carry the weights **40** up to 75% of the vertical distance back to the addition point **60** without expending any additional energy. The weights **40** may enter the addition position **64** from above, below, or from the sides. By way of example, one weight **40** may be added from the rear side, and the subsequent weight **40** may be added from the front side.

(36) The quantity, size, shape, and/or position of the weights **40** may vary between different embodiments. By way of example, the weights **40** may take up between 2% and 100% of the radius of the wheel **30**. By way of another example, the system **10** may include two sets of weights **40**, each set of weights being at a different distance D from the axis **34**. The weights **40** may extend across a large portion (e.g., most, all, etc.) of the width of the wheel **30**. By way of example, the weights **40** may have a width of 1 ft, 2 ft, 3 ft, 5 ft, 10 ft, 15 ft, 20 ft, or more.

(37) Although only one weight actuator **50** is shown in FIG. 2, in some embodiments, the system **10** includes multiple weight actuators **50**. By way of example, one weight actuator **50** may be positioned on each side of the wheel **30** to facilitate rapid movement of the weights **40** without the weight actuators **50** interfering with one another. In such embodiments, the width of the wheel **30** may be increased, and the weights **40** may be located at the center of the wheel **30** instead being of cantilevered off of one side of the wheel **30** to facilitate access from both sides of the wheel **30**.

(38) Referring to FIG. 7, the system **10** includes a control system **100** including processing circuitry, shown as controller **110**. The controller **110** includes a processor **112** and a memory device, shown as memory **114**. The memory **114** may contain one or more instructions that are executed by the processor **112** to control operation of the system **10**. The controller **110** is operatively coupled to the various components of the system **10**. The controller **110** configured to receive information from various components of the system **10** and/or provide information (e.g., commands) to various components of the system **10**.

(39) In some embodiments, the controller **110** controls the weight actuator **50**. Specifically, the controller **110** controls the weight actuator **50** to move the weights **40** from the subtraction position **60** to the addition position **64**. By way of example, the controller **110** may control the interface **52** to engage a weight **40** at the subtraction position **60**, control the lift **54** to raise the weight **40**, and control the interface **52** to disengage the weight **40** from the lift **54** and/or couple the weight **40** to the wheel **30**. The controller **110** may control the speed of the wheel **30** by varying the speed at which the weight actuator **50** moves the weights **40**.

(40) In some embodiments, the controller **110** controls the electric motor **72** and/or the batteries **76**. By way of example, the controller **110** may electrically couple the electric motor **72** to the batteries **76** while controlling the weight actuator **50** to drive the electric motor **72** such that the electric motor **72** provides electrical energy to charge the batteries **76**. By way of another example, the controller **110** may electrically couple the electric motor **72** to the batteries **76** and control the batteries **76** to power the electric motor **72**. By way of another example, the controller **110** may electrically couple the electric motor **72** and/or the batteries **76** to the loads **78**. The electric motor **72** and/or the batteries **76** may provide electrical energy to the loads **78**. In embodiments where the loads **78** includes an electrical grid, the electrical grid may provide electrical energy to power the electric motor **72** and/or charge the batteries **76**.

(41) In some embodiments, the control system **100** includes one or more sensors, shown as weight sensors **120**, that are operatively coupled to the controller **110**. The weight sensors **120** are configured to provide data indicative of a position, orientation, or status (e.g., a fill level) of one or more weights **40**. By way of example, the weight sensors **120** may include limit switches, proximity sensors, break beam sensors, or other sensors that provide data relating to a position of the weight **40**. By way of another example, the weight sensors **120** may include a rotation sensor (e.g., an optical encoder or potentiometer, etc.) that indicates an angular position of the wheel **30**. The controller **110** may utilize a predetermined relationship between the angular position of the wheel **30** and the positions of the weights **40** to determine the positions of the weights **40**.

(42) Referring to FIG. 8, a power transmission system, shown as system **200** is shown according to an exemplary embodiment. The system **200** illustrates how several of the system **10** may be combined with one another. As shown, the system **10** includes a first subassembly, shown as wheel assembly **210**. The wheel assembly **210** includes three of the wheels **30** arranged in series with one another. Specifically, three wheels **30** are each coupled to one output shaft **32**. In other

embodiments, the wheel assembly **210** includes more or fewer wheels **30** arranged in series with one another (e.g., 3, 5, 10, 20, 30, 50, 100, or more wheels **30**). The output shaft **32** is rotatably supported by a pair of bases **20**. The wheels **30** all rotate about a common axis **34** that extends through the output shaft **32**. Each wheel **30** is driven by a corresponding weight actuator **50** that repositions one or more weights **40** relative to the wheel **30**. The moment load of the weights **40** on each of the wheels **30** drives the output shaft **32**. The output shaft **32** is connected to an electric motor **72** by a belt **74**, such that all of the wheels **30** can drive and/or be driven by the electric motor **72**. By utilizing multiple wheels **30** to drive one output shaft **32**, the power output of the wheel assembly **210** may be increased relative to a system with only one wheel **30**. In other embodiments, the wheel assembly **210** includes more or fewer wheels **30**.

(43) The wheels **30** may be fixedly coupled to the output shaft **32**. Alternatively, the wheels **30** may be selectively coupled to the output shaft **32** by a clutch, ratchet, or one-way bearing. Such a configuration may permit the wheels **30** to rotate relative to one another. In some embodiments, a first wheel **30** may be coupled to the output shaft **32** after a second wheel **30** has already begun rotating, such that the momentum of the second wheel **30** imparts a torque on the first wheel **30**, facilitating rotation of the first wheel **30**. Such a configuration may be used to facilitate startup of rotation of the first wheel **30**.

(44) The system **200** further includes a second subassembly, shown as wheel assembly **220**. The wheel assembly **220** includes one wheel **30** is coupled to an output shaft **32**. The output shaft **32** is rotatably supported by a base **20**. The wheel **30** is directly coupled to the output shaft **32** such that the wheel **30** rotates about an axis **34**. The wheel **30** is driven by a weight actuator **50** that repositions one or more weights **40** relative to the wheel **30**. The moment load of the weights **40** on the wheel **30** drives the output shaft **32**. The output shaft **32** is connected to an electric motor **72** by a belt **74**, such that the wheel **30** can drive and/or be driven by the electric motor **72**.

(45) In some embodiments, the electric motors **72** of the wheel assembly **210** and the wheel assembly **220** are both electrically coupled to a common set of batteries **76** (e.g., selectively coupled as controlled by the controller **110**). Accordingly, the electric motors **72** may each receive electrical energy from the batteries **76** and/or supply electrical energy to the batteries **76**. In some embodiments, the electrical energy from one wheel assembly may be used to drive the other wheel assembly. By way of example, the wheel assembly **210** may be operating at a high speed while the wheel assembly **220** is stationary or operating at a low speed. The electric motor **72** of the wheel assembly **210** may provide electrical energy that is supplied to the electric motor **72** of the wheel assembly **220** to drive the wheel **30** of the wheel assembly **220**. Such a configuration may be used to facilitate startup of one of the wheel assemblies.

(46) Referring to FIG. **9**, a power transmission system, wheel assembly, power transmission assembly, power transmission, power conversion assembly, or generator, is shown as system **300**. The system **300** represents one possible arrangement of the generic system **10**. Accordingly, any description with reference to the system **10** may apply to the system **300**, except as otherwise stated.

(47) The weights **40** of the system **300** include a plurality of weights **302**. Each of the weights **302** forms a discrete piece that is selectively coupled to the wheel **30**. As shown, the system **300** includes eleven weights **302**. In other embodiments, the system **300** includes more or fewer weights **302**. As shown, the weights **302** are positioned along a circumference of the wheel **30**. In some embodiments, the weights **302** are rigid. In other embodiments, the weights **302** are flexible. By way of example, the weights **302** may bend into a concave shape when inserted into a track that extends along the circumference of the wheel **30**.

(48) The weights **302** may be shaped to facilitate densely packing the weights **302** with few gaps between the weights **302**. As shown, the weights **302** are shaped as trapezoidal prisms that extend along the axis **34**. Adjacent surface of the weights **302** engage one another, minimizing gaps between the weights **302** and maximizing the mass that can be supported by the wheel **30**. In other

embodiments, the weights **302** are otherwise shaped. By way of example, the weights **302** be shaped as triangular prisms that each have a triangular cross section. In such an embodiment, adjacent surfaces of the weights **302** may engage one another, similarly to the trapezoidal weights **302** of FIG. **9**. By way of another example, the weights **302** may have a rectangular cross section. In some such embodiments, the weights **302** are flat plates that can be packed closely to one another. By way of another example, the weights **302** may have circular or other shaped cross sections. In some embodiments, the system **300** includes weights having a variety of different shapes (e.g., weights **302** with triangular cross sections positioned between weights **302** with rectangular cross sections).

(49) The weights **302** may be selectively coupled to the wheel **30** such that the weights **302** may be decoupled from the wheel **30** when the weights **302** reach the subtraction position **60**. As shown, the system **300** includes a plurality of couplers **304** that selectively couple (e.g., selectively rotatably couple, selectively fixedly couple, etc.) one or more of the weights **302** to the wheel **30**. The controller **110** may control the couplers **304** to decouple the corresponding weights **302** from the wheel **30** when the weight **302** reaches the subtraction position **60**. The system **300** may include one coupler **304** that couples each of the weights **302** to the wheel **30**. In other embodiments, each coupler **304** couples multiple weights **302** to the wheel **30**. In some embodiments, the coupler **304** includes an electromagnet that may be activated to magnetically couple one or more weights **302** to the wheel **30**. In some embodiments, the coupler **304** includes a suction port coupled to a vacuum pump that applies suction to couple one or more weights to the wheel **30**. In some embodiments, the coupler **304** includes a clutch, latch, grabber, solenoid, or other device that mechanically engages one or more weights **302** to couple the one or more weights **302** to the wheel **30**.

(50) In some embodiments, the weights **302** are otherwise selectively coupled to the wheel **30**. By way of example, the system **300** may include a track that surrounds a portion of the wheel **30** extending between the addition position **64** and the subtraction position **60**. A weight **302** may be dropped into a recess defined by the wheel **30** at the addition position **64**. The recess may extend radially outward such that the weight **302** is captured within the recess and prevented from moving along the circumference of the wheel. As the wheel **30** rotates, the wheel **30** may force the weight **302** into engagement with the track, which prevents the weight **302** from exiting the recess. The track may end immediately before the subtraction position **60** such that the weight **302** is permitted to exit the recess at the subtraction position **60** due to the force of gravity.

(51) Referring still to FIG. **9**, the weight actuator **50** includes a movable structure or rotating structure, shown as arm **310**. The arm **310** is coupled to at least one actuator or lift, shown as motor **312**, that is configured to control motion of the arm **310**. As shown, the motor **312** controls rotation of the arm **310** about an axis of rotation that extends substantially parallel to the axis **34**. Additionally or alternatively, the motor **312** may control translational movement, articulation (e.g., bending), and/or telescoping (i.e., a change in length) of the arm **310**. Positioned at the end of the arm **310** is an interface (e.g., an electromagnet, a claw or grabber, a vacuum, etc.), shown as end effector **314**, that is configured to selectively couple the weight **302** to the arm **310**. In operation, the end effector **314** engages each weight **302** after the weight **302** has reached the subtraction position **60**. The motor **312** then causes the arm **310** to raise the weight **302** to an elevated position. With the weight **302** in the elevated position, the end effector **314** disengages the weight **302** from the arm **310**, permitting the weight **302** to be coupled to the wheel **30** at the addition position **64** (e.g., by the coupler **304**). In some embodiments, the arm **310** and/or the end effector **314** are configured to rotate the weights **302** to facilitate depositing the weights **302** in the alignment position **64** with the weights **302** in a desired orientation.

(52) As shown in FIG. **9**, the system **300** includes an electrical energy generator, energy capturer, or recovery platform, shown as recovery mat **320**. The recovery mat **320** is positioned below the subtraction position **60** such that the weights **302** fall onto the recovery mat **320** after decoupling from the wheel **30**. Accordingly, the recovery mat **320** experiences the impact force from the falling

weight **302**. The recovery mat **320** compresses or deflects below the weight **302**, receiving the kinetic energy from the weight **302** and slowing the weight **302** to a stop. The weight actuator **50** may then recover the weight **302** from atop the recovery mat **320** and move the weight **302** to the elevated position. In other embodiments, the recovery mat **320** is omitted.

(53) The recovery mat **320** may be configured to convert a portion of the kinetic energy from the falling weight **302** to electrical energy. By way of example, the recovery mat **320** may include a platform positioned along a top surface of the recovery mat **320** that is configured to move vertically. The platform may be coupled to one or more electric motors or generators, such that movement of the platform drives the electric motors to provide electrical energy. The electrical energy provided by the electric motors may be used to power other components of the system **300** (e.g., to drive the weight actuator **50**, to charge the batteries **76**, etc.). The force of the weight **302** may drive the platform downward, thereby driving the electric motors. After the weight **302** has been removed from the recovery mat **320**, the motors may drive the platform upward to reset the position of the platform. In one example, the platform is supported by one or more linear actuators, and the linear actuators are coupled to the motor. Accordingly, the linear actuators may couple the movement of the platform to the movement of the electric motors.

(54) Referring still to FIG. **9**, the system **300** further includes a holding bay, holding area, hopper, or staging assembly, shown as hopper assembly **350**. The hopper assembly **350** temporarily stores the weights **302** after the weights **302** are raised by the weight actuator **50** until the weights **302** are coupled to the wheel **30**. The hopper assembly **350** facilitates immediate introduction of the weights **302** to the wheel **30** regardless of the position of the arm **310**. As shown, the hopper assembly **350** includes a hopper body **352** that defines a passage **354** that receives and contains one or more of the weights **302**. The weight actuator **50** deposits the weights **302** at a first end of the passage **354**, and the weights **302** exit an opposing, second end of the passage **354** to be introduced back to the wheel **30** at the addition position **64**. An actuator assembly, shown as hopper index actuator **356**, is coupled to the hopper body **352** adjacent the second end of the passage **354**. The hopper index actuator **356** includes an index wheel that engages the weights **302** within the passage **354** that is driven by an electric motor. With the hopper index actuator **356** stationary, the hopper index actuator **356** prevents the weights **302** from exiting the passage **354**. The hopper index actuator **356** may be driven to dispense a weight **302** onto the wheel **30**. In other embodiments, the hopper assembly **350** is omitted, and the weight actuator **50** moves the weights **302** directly onto the wheel **30** at the addition position **64**.

(55) In some embodiments, the arm **310** is configured to move along a path that utilizes the momentum of the weights **302** caused by motion of the wheel **30**. By way of example, a weight **302** at the subtraction position **60** may have momentum that would continue to carry the weight **302** horizontally away from the wheel **30**. The path of the arm **310** after the end effector **314** engages the weight **302** may initially move the weight **302** in the first direction, such that the momentum **302** of the weight **302** moves the weight **302** along the path without any energy being consumed by the motor **312**. When the momentum has been depleted, the motor **312** may begin to drive the arm **310** to complete the movement of the weight **302** to the hopper assembly **350** or the addition position **64**.

(56) Referring to FIG. **10**, a power transmission system, wheel assembly, power transmission assembly, power transmission, power conversion assembly, or generator, is shown as system **400**. The system **400** represents one possible arrangement of the generic system **10**. Accordingly, any description with reference to the system **10** may apply to the system **400**, except as otherwise stated.

(57) The system **400** includes a weight **402**. The weight **402** is a discrete piece that is slidably coupled to the wheel **30**. Specifically, the system **400** includes a guide, slide, or track, shown as track **404**, that is coupled (e.g., fixedly coupled) to the wheel **30**. The track **404** is slidably coupled to the weight **402** (e.g., by one or more bushings or bearings) such that the weight **402** moves along

a path P defined by the track **404**. As shown, the path P is centered about the track **404** and extends along the track **404**. Accordingly, the weight **402** is selectively repositionable relative to the wheel **30** about along the track **404**.

(58) The weight actuator **50** of the system **400** includes an actuator (e.g., an electric motor, an electric linear actuator, etc.), shown as weight actuator **406**, that is configured to move the weight **402** along the track **404**. The weight actuator **406** is coupled to the wheel **30** and the weight **402**. By way of example, the weight actuator **406** may include a linear actuator that is coupled to the weight **402** and configured to reposition the weight **402** along the track **404**. By way of another example, the weight actuator **406** may drive a belt or a rack gear that is fixedly coupled to the weight **402** such that the weight **402**, thereby driving the weight **402** along the track **404**.

(59) As shown in FIG. **10**, the track **404** extends perpendicular to the output shaft **32**, and the track **404** is aligned with the output shaft **32**. Accordingly, the path P of the weight **402** intersects the axis **34** of the wheel **30**. The weight **402** travels radially relative to the axis **34** as the weight **402** moves along the path P. Accordingly, as the weight **402** travels along the path P, the distance between the weight **402** and the axis **34** varies, changing the moment of inertia of the weight **402** about the axis **34**. As shown, the track **404** extends on both sides of the output shaft **32**. Accordingly, as the weight **402** moves along the track **404**, the moment of inertia of the weight **402** decreases as the weight **402** moves toward the axis **34**, then increases as the weight **402** moves beyond the axis **34**. In other embodiments, the path of the weight **402** extends on only one side of the axis **34**.

(60) In operation, the weight **402** may begin in the addition position **64** with the weight **402** located at a first end of the path P, such that the gravitational force on the weight **402** drives the weight **402** and the wheel **30** to rotate about the axis **34** (e.g., counterclockwise as shown in FIG. **10**). When the weight **402** reaches the subtraction position **60** (e.g., reaches a predetermined positions, reaches a predetermined range of positions, etc.), the controller **110** activates the weight actuator **406** to move the weight **402** along the track **404** toward the axis **34**. This reduces the moment load of the force of gravity on the weight **402** that would oppose counterclockwise movement of the wheel **30**. In embodiments where the path P and the track **404** extend on both sides of the axis **34**, such as the embodiment of FIG. **10**, the controller **110** may control the weight actuator **406** to continue moving the weight **402** beyond the axis **34**, toward the second end of the path P, where the gravitational force on the weight **402** again drives the weight **402** and the wheel **30** to rotate counterclockwise. In embodiments where the path P and the track **404** extends on only one side of the axis **34**, the controller **110** may control the weight actuator **406** to retain the weight **402** as close to the axis **34** possible (e.g., near the axis **34**, at the axis **34**). In this position, the moment load of the force of gravity on the weight **402** that would oppose counterclockwise movement of the wheel **30** is minimized as the wheel **30** continues to rotate (e.g., due to momentum, due to the effect of other weights **402**, due to an input from the electric motor **72**, etc.). Once the wheel **30** has rotated sufficiently for the first end of the path P to approach (e.g., fall within a threshold range of positions relative to) the addition position **64**, the controller **110** may control the weight actuator **406** to return the weight **402** to the first end of the path P, where the gravitational force on the weight **402** again drives the wheel **30** to rotate clockwise.

(61) In other embodiments, the system **400** includes multiple weights **402**. By way of example, the system **400** may include multiple tracks **404**, each track **404** supporting a different weight **402**. The tracks **404** may have differing lengths. By way of example, the system **400** may include one track **404** that extends across the axis **34** and two additional tracks **404** that extend on only one side of the axis **34**. The movement of each weight **402** may be controlled individually by a corresponding weight actuator **406**.

(62) Referring to FIG. **11**, a power transmission system, wheel assembly, power transmission assembly, power transmission, power conversion assembly, or generator, is shown as system **500**. The system **500** represents one possible arrangement of the generic system **10**. Accordingly, any description with reference to the system **10** may apply to the system **500**, except as otherwise

stated.

(63) The weights **40** of the system **500** include volumes of flowable material that are held within containers coupled to the wheel **30**. As used herein, the term “flowable material” can refer to any collection of molecules, particles, elements, or objects that are capable of flowing into a container. A flowable material can include liquids, solids, or a combination thereof. By way of example, the flowable material may include materials that are liquid at room temperature and atmospheric pressure, such as water, mercury, oil, or alcohol. The flowable material may include bulk solids, granules, grains, balls, pellets, fragments, pieces, powders, or other flowable forms of materials that are solid at room temperature and atmospheric pressure. By way of example, the flowable material may include sand, soil, gravel, metal balls or powders (e.g., lead pellets, steel ball bearings, steel cubes, iron powder, etc.), salt, grains (e.g., corn, rice, etc.), sugar, plastic pellets, or glass beads.

(64) Referring still to FIG. **11**, the system **500** includes a plurality of containers, compartments, or vessels (e.g., buckets, cups, tanks, etc.), shown as buckets **502** that are coupled to the wheel **30**. Although the buckets **502** are shown as being square, the buckets **502** may be otherwise shaped, (e.g., triangular, rectangular, spherical, cylindrical, etc.). Each bucket **502** defines a volume or recess. As shown, each bucket **502** is rotatably coupled to the wheel **30** such that the buckets **502** maintain a substantially constant orientation relative to the direction of gravity as the wheel **30** rotates. As shown, coupled to the bottom of each bucket **502** is a valve or release, shown as a pair of doors **504**, that are selectively repositionable relative to the bucket **502**. In a closed configuration (e.g., as shown in the uppermost bucket **502** of FIG. **11**), the doors **504** seal against the bucket **502**, preventing flowable material from exiting the bucket **502**. In an open configuration (e.g., as shown in the lowermost bucket **502** of FIG. **11**), the doors **504** open to permit the flowable material to leave the bucket **502** through an aperture defined at the bottom of the bucket **502**.

(65) During operation, the buckets **502** are filled with a volume **506** of flowable material when in the addition position **64**. In some embodiments, the volume **506** is a predetermined amount of flowable material. The weight of the volume **506** of flowable material forces the bucket **502** downward, driving rotation of the wheel **30** (e.g., counterclockwise as shown in FIG. **11**). When the bucket **502** reaches the subtraction position **60**, an actuator (e.g., a latch, one or more linear actuators, etc.), shown as door actuator **508**, moves the doors **504** of the bucket **502** to the open configuration, permitting the volume **506** of flowable material to fall into a container, vessel, holding bay, or tank, shown as bottom tank **510**. In some embodiments, the door actuator **508** is operated by the controller **110** (e.g., using electronic signals). In other embodiments, the door actuator **508** is mechanically actuated (e.g., the door actuator **508** engages a protrusion coupled to the base **20** when the wheel **30** reaches a predetermined position, causing the door actuator **508** to open the doors **504**). In some embodiments, the bottom tank **510** is offset to the left as shown in FIG. **11** to ensure that any flowable material dispensed prior to the bucket **502** entering the subtraction position **60** is captured. After the volume **506** has been emptied from the bucket **502**, the door actuator **508** closes the doors **504**, and the bucket **502** moves back toward the addition position **64**. Alternatively, the door actuator **508** may open the doors **504** while the bucket **502** is between the subtraction position **60** and the addition position **64**, or while the bucket **502** is at the addition position **64**.

(66) In some embodiments, each bucket **502** includes an actuated pushing mechanism (e.g., a plunger) that actively forces the flowable material out of the bucket **502** and into the bottom tank **510**. The pushing mechanism may dispense the flowable material more quickly than gravity, which may facilitate faster rotation speeds of the wheel **30**. The pushing mechanism may also be configured to displace a consistent amount of flowable material each time (e.g., by controlling the stroke length of the pushing mechanism). In some embodiments, the amount of flowable material dispensed is configured to completely empty the bucket **502**.

(67) The bottom tank **510** contains a volume **512** of flowable material. The volume **512** may be larger than the volume **506**. In the system **500**, the weight actuator **50** includes an elevator, lift, or

actuator, shown as pump **520**, that is fluidly coupled to the bottom tank **510** and to a conduit (e.g., a pipe, tube, hose, etc.), shown as pipe **522**. The pump **520** is configured to drive a portion of the flowable material from the volume **512** within the bottom tank **510**, through the pipe **522**, to a container, vessel, holding bay, or tank, shown as top tank **524**, that is fluidly coupled to the pipe **522**. The pump **520** may be powered by electrical energy from the electric motor **72** and/or the battery **76**. In some embodiments, the system **500** utilizes multiple pumps **520** (e.g., in series, in parallel).

(68) The top tank **524** contains a volume **526** of the flowable material. The top tank **524** is fluidly coupled to a dispenser, shown as fill valve **528**. The fill valve **528** is configured to control a flow of material from the top tank **524** into the buckets **502**. In some embodiments, the fill valve **528** is configured to dispense a predetermined amount (e.g., a predetermined volume, a predetermined mass, etc.) of the flowable material into each bucket **502**. In some embodiments, the fill valve **528** forms a seal with each bucket **502** when the bucket **502** is in the addition position **64** to minimize spillage of the flowable material while the bucket **502** is being filled. By way of example, the bucket **502** may include a door that opens when the fill valve **528** is in close proximity to the bucket **502**, and the fill valve **528** may form a seal with the door. In some embodiments, the fill valve **528** includes an actuated pushing mechanism (e.g., a plunger) that actively forces the flowable material into the bucket **502**. The pushing mechanism may dispense the flowable material more quickly than gravity, which may facilitate faster rotation speeds of the wheel **30**. The pushing mechanism may also be configured to displace a consistent amount of flowable material each time (e.g., by controlling the stroke length of the pushing mechanism). In some embodiments, the amount of flowable material dispensed is configured to completely fill each bucket **502**.

(69) In other embodiments, the buckets **502** are fixedly coupled to the wheel **30**, such that the buckets **502** rotate with the wheel **30**. In such embodiments, the flowable material may automatically be poured from the buckets **502** when the buckets **502** near the subtraction position **60**, as the rotation of the wheel **30** inverts the buckets **502**. In such embodiments, the doors **504** and/or the door actuators **508** may be omitted. In other embodiments, another type of elevator is utilized to move the flowable material from the bottom tank **510** to the top tank **524**. By way of example, the elevator may include one or more buckets or containers coupled to a conveyor. The conveyor drives the buckets along a path such that the buckets scoop the flowable material from the bottom tank **510** and pour the flowable material into the top tank **524**.

(70) In other embodiments, the system **500** includes a conveyor or elevator (e.g., driven by pulleys and/or gears) that exchanges the bottom tank **510** with the top tank **524** after the bottom tank **510** is filled. By way of example, the conveyor may raise the bottom tank **510** above the wheel **30** such that the fill valve **528** dispenses flowable material from the bottom tank **510**. By way of another example, the conveyor may lower the top tank **524** below the wheel **30** to receive the flowable material from the buckets **502**. In some embodiments, additional tanks are introduced into this rotation to form a queue of tanks that are ready to be swapped into position above or below the wheel **30**. This may reduce any potential spillage of flowable material while the tanks are being exchanged.

(71) Referring to FIGS. **12** and **13**, a power transmission system, wheel assembly, power transmission assembly, power transmission, power conversion assembly, speed governor, or energy storage device, generator, is shown as system **600**, according to an exemplary embodiment. The system **600** may be substantially similar to the system **10**, except as otherwise specified herein.

(72) The system **10** includes a chassis, frame, or base assembly, shown as base **602**, configured to support the other components of the system **600**. The base **602** defines an aperture that receives a shaft support shaft, shown as output shaft **604**. A series of wheels, rotating frames, or rotating support structures, shown as wheels **606**, are coupled (e.g., fixedly coupled, selectively coupled by one or more clutches, etc.) to the output shaft **604**. The output shaft **604** and the wheels **606** rotate about an axis of rotation, shown as axis **608**. The axis **608** extends along the output shaft **604** (e.g.,

is centered about the output shaft **604**). In some embodiments, the axis **608** extends substantially vertically. In such embodiments, the wheels **606** each rotate within a substantially horizontal plane (e.g., a plane perpendicular to the axis **608**). In other embodiments, the axis **608** is otherwise oriented. By way of example, the axis **608** may extend substantially horizontally. In such embodiments, the wheels **606** each rotate within a substantially vertical plane (e.g., a plane perpendicular to the axis **608**).

(73) The output shaft **604** is coupled to an electromagnetic device or motor/generator, shown as electric motor **610**. The electric motor **610** is electrically coupled to an energy storage device (e.g., batteries, capacitors, etc.), shown as batteries **612**. In some configurations, the output shaft **604** drives the electric motor **610** to produce electrical energy that is stored within the batteries **612**. In other configurations, the electric motor **610** receives electrical energy from batteries **612** and drives the output shaft **604**. The output shaft **602** is further coupled to a speed sensor or rotation sensor (e.g., an optical encoder), shown as rotation sensor **630**. The rotation sensor **630** is configured to provide sensor data indicative of a rotational speed of the output shaft **602** and/or the wheels **606**.

(74) The system **600** includes a series of sliding weight assemblies **620** coupled to each wheel **606**. FIG. 12 illustrates the arrangement of the sliding weight assemblies **620** on the uppermost wheel **606**, although it should be understood that each of the wheels **606** may have a similar arrangement. Each sliding weight assembly **620** includes a weight **622** that is slidably coupled to a guide, shown as track **624**. The tracks **624** are each coupled to one of the wheels **606**. The tracks **624** are each oriented extending radially outward relative to the axis **608**. The weights **622** are repositionable along the length of each track **624** between an outermost position (shown in solid lines in FIG. 12) and an innermost position (shown in dashed lines in FIG. 12). The weights **622** are moved along the tracks **624** by an actuators (e.g., an electric motor driving a belt or rack gear, an electric linear actuator, etc.), shown as weight actuators **626**. The weight actuators **626** may be powered by the batteries **612**. Operation of the electric motor **610**, the batteries **612**, and the weight actuators **626** may be controlled by a controller (e.g., the controller **110**). The controller may control these functions based on sensor data from the rotation sensor **630**.

(75) During operation, the weights **622** begin in their outermost positions, and the wheels **606** are driven to a starting speed by the electric motor **610**, at which point the system **600** contains an initial amount of kinetic energy. The kinetic energy may be used to perform one or more functions (e.g., to drive a PTO similar to the PTO **70**, to drive the electric motor **610** to generate electric energy, due to frictional losses, etc.). As the kinetic energy in the system **600** decreases, the wheels **606** begin to rotate more slowly. In response to a decrease in speed of the wheels **606** (e.g., as sensed by the rotation sensor **630**), the weight actuators **626** drive the weights **622** toward the innermost positions. This decreases the moment of inertia of the system **600**, increasing the speed of the wheels **606** until a desired speed (e.g., the starting speed). A controller may repeat this process until all of the weights **622** are at the innermost positions. At this point, the electric motor **610** may drive the wheels **606** to maintain the desired speed, and the weight actuators **626** may return the weights **622** to the outermost positions.

(76) In some embodiments, all of the weights **622** move toward the axis **608** in unison. In other embodiments, a first subset of the weights **622** initially move toward the axis **608**, and a second subset of the weights **622** begin moving toward the axis **608** after the first subset reach the innermost positions. In some embodiments, each sliding weight assembly **620** includes multiple weights **622** positioned on each track **624** (e.g., 2, 3, 4, 5, 6, 7, or more weights **622**). Once one of the weights **622** reaches the innermost position, the next weight **622** may begin moving toward the axis **608**. This process may repeat for each weight **622**.

(77) In some embodiments, each wheel **606** has a relatively large diameter to facilitate an extended travel of the weights **622**. By way of example, the wheels **606** may be 25 ft, 50 ft, 100 ft, 1000 ft, 5000 ft or greater in diameter. The heights of the wheels **606** may be relatively small to facilitate stacking multiple wheels **606**. In some embodiments, the system **600** includes more or fewer

wheels **606** (e.g., 1, 2, 4, 5, 6, 7, 8, 9, 10, 15, 20, 30, 50, 100, or more wheels **606**). In some embodiments, the output shaft **604** is coupled to multiple electric motors **610**. In some embodiments, the output shaft **604** is coupled to the electric motors **610** through a power transmission (e.g., a gear train).

(78) In some embodiments, it is advantageous to minimize the weight of the wheels **606** and the output shaft **604** and maximize the weight of the weights **622**. This arrangement maximizes the amount of output power provided by the system **600** for weights **622** of a given size while minimizing the energy required to accelerate the wheels **606**. In some embodiments, the wheel **606** includes relatively lightweight materials, such as aluminum or bamboo. In some embodiments, the weights **622** include relatively dense materials, such as lead, mercury, steel, or iron.

(79) In some embodiments, the tracks **624** and/or the weights **622** include a ratchet assembly that selectively prevents outward movement of the weights **622**. Such a ratchet assembly may apply an inward force on the weights **622** to limit outward movement of the weights **622** during rotation. The ratchet assembly may be released to permit the weights **622** to return to their outermost positions (e.g., automatically due to rotation of the system **600**, without the application of any energy by the weight actuators **626**).

(80) As utilized herein, the terms “approximately,” “about,” “substantially,” and similar terms are intended to have a broad meaning in harmony with the common and accepted usage by those of ordinary skill in the art to which the subject matter of this disclosure pertains. It should be understood by those of skill in the art who review this disclosure that these terms are intended to allow a description of certain features described and claimed without restricting the scope of these features to the precise numerical ranges provided. Accordingly, these terms should be interpreted as indicating that insubstantial or inconsequential modifications or alterations of the subject matter described and claimed are considered to be within the scope of the disclosure as recited in the appended claims.

(81) It should be noted that the term “exemplary” and variations thereof, as used herein to describe various embodiments, are intended to indicate that such embodiments are possible examples, representations, or illustrations of possible embodiments (and such terms are not intended to connote that such embodiments are necessarily extraordinary or superlative examples).

(82) The term “coupled” and variations thereof, as used herein, means the joining of two members directly or indirectly to one another. Such joining may be stationary (e.g., permanent or fixed) or moveable (e.g., removable or releasable). Such joining may be achieved with the two members coupled directly to each other, with the two members coupled to each other using a separate intervening member and any additional intermediate members coupled with one another, or with the two members coupled to each other using an intervening member that is integrally formed as a single unitary body with one of the two members. If “coupled” or variations thereof are modified by an additional term (e.g., directly coupled), the generic definition of “coupled” provided above is modified by the plain language meaning of the additional term (e.g., “directly coupled” means the joining of two members without any separate intervening member), resulting in a narrower definition than the generic definition of “coupled” provided above. Such coupling may be mechanical, electrical, or fluidic.

(83) References herein to the positions of elements (e.g., “top,” “bottom,” “above,” “below”) are merely used to describe the orientation of various elements in the FIGURES. It should be noted that the orientation of various elements may differ according to other exemplary embodiments, and that such variations are intended to be encompassed by the present disclosure.

(84) The hardware and data processing components used to implement the various processes, operations, illustrative logics, logical blocks, modules and circuits described in connection with the embodiments disclosed herein may be implemented or performed with a general purpose single- or multi-chip processor, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), or other programmable logic device, discrete

gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general purpose processor may be a microprocessor, or, any conventional processor, controller, microcontroller, or state machine. A processor also may be implemented as a combination of computing devices, such as a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration. In some embodiments, particular processes and methods may be performed by circuitry that is specific to a given function. The memory (e.g., memory, memory unit, storage device) may include one or more devices (e.g., RAM, ROM, Flash memory, hard disk storage) for storing data and/or computer code for completing or facilitating the various processes, layers and modules described in the present disclosure. The memory may be or include volatile memory or non-volatile memory, and may include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described in the present disclosure. According to an exemplary embodiment, the memory is communicably connected to the processor via a processing circuit and includes computer code for executing (e.g., by the processing circuit or the processor) the one or more processes described herein.

(85) The present disclosure contemplates methods, systems and program products on any machine-readable media for accomplishing various operations. The embodiments of the present disclosure may be implemented using existing computer processors, or by a special purpose computer processor for an appropriate system, incorporated for this or another purpose, or by a hardwired system. Embodiments within the scope of the present disclosure include program products comprising machine-readable media for carrying or having machine-executable instructions or data structures stored thereon. Such machine-readable media can be any available media that can be accessed by a general purpose or special purpose computer or other machine with a processor. By way of example, such machine-readable media can comprise RAM, ROM, EPROM, EEPROM, or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code in the form of machine-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer or other machine with a processor. Combinations of the above are also included within the scope of machine-readable media. Machine-executable instructions include, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machines to perform a certain function or group of functions.

(86) Although the figures and description may illustrate a specific order of method steps, the order of such steps may differ from what is depicted and described, unless specified differently above. Also, two or more steps may be performed concurrently or with partial concurrence, unless specified differently above. Such variation may depend, for example, on the software and hardware systems chosen and on designer choice. All such variations are within the scope of the disclosure. Likewise, software implementations of the described methods could be accomplished with standard programming techniques with rule-based logic and other logic to accomplish the various connection steps, processing steps, comparison steps, and decision steps.

(87) It is important to note that the construction and arrangement of the system **10** as shown in the various exemplary embodiments is illustrative only. Additionally, any element disclosed in one embodiment may be incorporated or utilized with any other embodiment disclosed herein. For example, the wheel assembly **210** of FIG. **8** may include both the system **300** of FIG. **9** and the system **400** of FIG. **10**. Although only one example of an element from one embodiment that can be incorporated or utilized in another embodiment has been described above, it should be appreciated that other elements of the various embodiments may be incorporated or utilized with any of the other embodiments disclosed herein.

Claims

1. A power transmission assembly, comprising: a base; a rotating frame rotatably coupled to the base and configured to rotate about an axis; an output shaft coupled to the rotating frame; a weight selectively repositionable relative to the rotating frame; and a weight actuator configured to reposition the weight relative to the rotating frame to move the weight from a subtraction position located at a first height to an addition position located at a second height, wherein the subtraction position and the addition position are the same distance from the axis, wherein the second height is greater than the first height such that a gravitational force on the weight drives the rotating frame to rotate about the axis and drive the output shaft, wherein the weight actuator is configured to receive energy from an external energy source.
2. The power transmission assembly of claim 1, wherein the weight is a first weight, further comprising a second weight selectively repositionable relative to the rotating frame, wherein the weight actuator is configured to reposition the second weight relative to the rotating frame from the subtraction position to the addition position.
3. The power transmission assembly of claim 2, further comprising a controller operatively coupled to the weight actuator and configured to control the weight actuator to reposition the second weight relative to the rotating frame from the subtraction position to the addition position while the gravitational force on the first weight drives the rotating frame to rotate about the axis.
4. The power transmission assembly of claim 3, wherein the axis is a substantially horizontal axis.
5. The power transmission assembly of claim 1, further comprising an electromagnetic device coupled to the output shaft, wherein the electromagnetic device is configured to receive rotational mechanical energy from the output shaft and generate electrical energy.
6. The power transmission assembly of claim 5, further comprising: an energy storage device configured to store electrical energy; and a controller configured to electrically couple the energy storage device to the electromagnetic device such that the electromagnetic device drives rotation of the rotating frame in response to a startup request for the power transmission assembly.
7. The power transmission assembly of claim 1, wherein the weight is selectively coupled to the rotating frame, and wherein the weight is decoupled from the rotating frame in response to the weight entering the subtraction position.
8. The power transmission assembly of claim 7, wherein the weight is a volume of flowable material, further comprising a container coupled to the rotating frame and configured to receive the volume of flowable material.
9. The power transmission assembly of claim 8, wherein the flowable material includes a liquid.
10. The power transmission assembly of claim 8, wherein the flowable material includes a solid.
11. The power transmission assembly of claim 7, further comprising a hopper assembly configured to contain the weight, wherein the weight actuator is configured to reposition the weight relative to the rotating frame from the subtraction position to the hopper assembly, and wherein the hopper assembly is configured to dispense the weight onto the rotating frame at the addition position.
12. The power transmission assembly of claim 1, further comprising a track coupled to the rotating frame, wherein the weight is slidably coupled to the track such that the weight is selectively repositionable relative to the rotating frame along a path defined by the track.
13. The power transmission assembly of claim 12, wherein the axis intersects the path such that the weight is repositionable to a position in which the axis intersects the weight.
14. The power transmission assembly of claim 1, wherein the rotating frame is a first rotating frame, the weight is a first weight, the weight actuator is a first weight actuator, further comprising: a second rotating frame rotatably coupled to the base and configured to rotate about the axis; a second weight selectively repositionable relative to the second rotating frame; and a second weight actuator configured to reposition the second weight relative to the second rotating frame such that a

gravitational force on the second weight drives the second rotating frame to rotate about the axis and drive the output shaft.

15. The power transmission assembly of claim 1, wherein the base is a first base, the rotating frame is a first rotating frame, the axis is a first axis, the output shaft is a first output shaft, the weight is a first weight, the weight actuator is a first weight actuator, further comprising: a second base; a second rotating frame rotatably coupled to the second base and configured to rotate about a second axis; a second output shaft coupled to the second rotating frame; a second weight selectively repositionable relative to the second rotating frame; a second weight actuator configured to reposition the second weight relative to the second rotating frame such that a gravitational force on the second weight drives the second rotating frame to rotate about the second axis and drive the output shaft; a first electromagnetic device coupled to the first output shaft; and a second electromagnetic device coupled to the second output shaft, wherein the first electromagnetic device is configured to provide electrical energy to the second electromagnetic device such that the second electromagnetic device drives the second rotating frame.

16. A power transmission assembly, comprising: a base; a rotating frame rotatably coupled to the base and configured to rotate about an axis; an output shaft coupled to the rotating frame; a weight slidably coupled to the rotating frame; a weight actuator configured to move the weight between (a) an inner position located at a first distance from the axis and (b) an outer position located at a second distance from the axis, the second distance being greater than the first distance; an electric motor driven by a rotational motion of the output shaft about the axis; and a controller operatively coupled to the weight actuator and configured to control the weight actuator to move the weight from the outer position to the inner position in response to a decrease in rotational speed of the rotating frame.

17. The power transmission assembly of claim 1, wherein the external energy source comprises an electric grid.

18. The power transmission assembly of claim 1, wherein the external energy source comprises a battery.
